

**Quiz No. 3****CS3004****Software Design and Analysis****Total marks: 30****Total time: 45 minutes****Date: 28/04/2026**

Reg. No. _____

Name. _____

Scenario: Smart Home Automation System

A smart home automation system allows users to control lights, thermostats, and security cameras through a central mobile application. The system communicates with devices using a common Device Control Interface. To handle unreliable internet connectivity, a Local Hub Component is installed inside the home. The mobile app sends commands to the Local Hub when internet access is available, and the hub forwards these commands to smart devices over the local network. Therefore, the Local Hub both provides the Device Control Interface (to receive commands) and requires it (to send commands to devices). The backend cloud system includes a Command Processor Component, which authenticates and executes user commands, and an Automation Engine Component, which generates automated commands based on schedules (e.g., “turn off lights at 10 PM”). The Automation Engine retrieves user preferences from a User Settings Component to personalize automation. All commands generated by the Automation Engine are sent to the Command Processor for execution. A Logging Component records all system events and device commands. Both the Command Processor and Automation Engine use this logging service.

- 1) List four major components from the scenario and write one core responsibility for each.
- 2) Assign the most appropriate UML stereotype to each component identified in Q1. (e.g., <<application>>, <<database>>, <<library>>, <<executable>>, <<web service>>)
- 3) For any three components, identify:
 - a. One provided interface (name it and state what service it offers).
 - b. One required interface (name it and state what service it needs)
- 4) Relationships Between Components
 - a. Describe one assembly connector between two components and its purpose.
 - b. Identify two dependency (<<use>>) relationships and specify the components involved in each.
- 5) Draw one complete UML Component Diagram for the Smart Home Automation System described in the scenario. Your component diagram must include:
 - a. At least 5 components (labeled with <<component>> stereotype)
 - b. Provided interfaces using lollipop (circle) notation
 - c. Required interfaces using socket (half-circle) notation
 - d. The Device Control Interface clearly shown and used by at least two different components
 - e. Proper assembly connectors
 - f. At least 2 dependency arrows (<<use>> dashed arrows)

SOLUTION

Q1:

Component	Core Responsibility
Local Hub Component	Receives commands from the mobile app and forwards them to smart devices on the local network.
Command Processor Component	Authenticates and executes user/device commands.
Automation Engine Component	Generates automated commands based on schedules and user preferences.
Logging Component	Records all system events and command activities.

Q2:

Component	UML Stereotype
Local Hub Component	<<executable>>
Command Processor Component	<<application>>
Automation Engine Component	<<application>>
Logging Component	<<library>>
User Settings Component	<<database>>

Q3: Provided and Required Interfaces (Any Three Components)

1. Local Hub Component

- **Provided Interface:** IDeviceControl
 - Service: Receives device control commands from the mobile app/cloud.
- **Required Interface:** IDeviceControl
 - Service Needed: Sends commands to smart devices.

2. Automation Engine Component

- **Provided Interface:** IAutomationService
 - Service: Generates scheduled automation commands.
- **Required Interface:** IUserSettings
 - Service Needed: Retrieves user preferences/settings.

3. Command Processor Component

- **Provided Interface:** ICommandExecution
 - Service: Authenticates and executes commands.
- **Required Interface:** ILoggingService
 - Service Needed: Stores logs/events in the logging system.

Q4 A: One Assembly Connector

Between:

Automation Engine ↔ Command Processor

Purpose:

The Automation Engine sends automatically generated commands to the Command Processor for authentication and execution.

B: Two Dependency (<<use>>) Relationships

Automation Engine <<use>> Logging Component

- Used for recording automation events.

Command Processor <<use>> Logging Component

- Used for recording command execution logs.

Q5:

